



QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-2

Course Code: CSE 3004

Set number: 3

Duration: 90 min

Course Title: Design & Analysis of Algorithm

Date of Exam: 03/04/2023 (An)

Total Marks: 50

(G2)

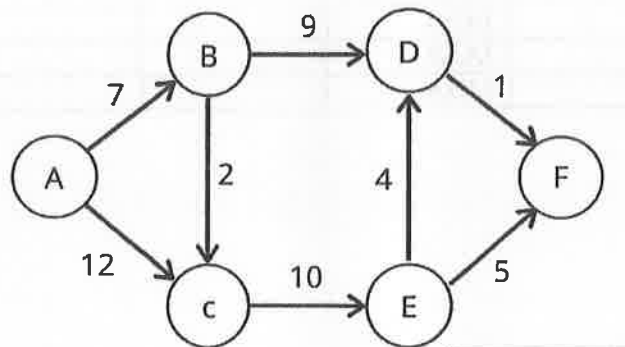
Q1. How optimal parenthesization of a matrix-chain product helps to improve the efficiency. Justify the answer with given input. $\langle 4; 10; 3; 9; 5; 20; 6 \rangle$. **(10 Marks)**

Q2. What is optimal binary search tree. Generate the optimal binary search tree for the following elements using dynamic programming approach.

Elements: $\langle 10, 20, 25, 30 \rangle$ and their corresponding frequency: $\langle 3, 2, 5, 4 \rangle$

(10 Marks)

Q3. In Dijkstra algorithm why do we use a priority queue as opposed to a regular first in first out queue? Solve the given example on which Dijkstra algorithm fails to compute shortest paths if a first in first out queue is used. **(10 Marks)**



Q4. Maximize the equation using simplex method. Where, Maximize $Z = 5x_1 + 3x_2$

Subject to

$$x_1 + x_2 \leq 2$$

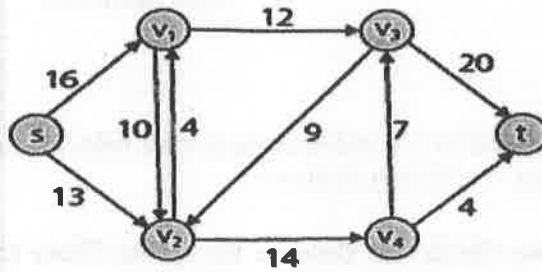
$$5x_1 + 8x_2 \leq 10$$

$$3x_1 + 8x_2 \leq 12.$$

Where $x_1, x_2 \geq 0$

(10 Marks)

Q5. What is the maximum flow possible in the below mentioned graph from source node S to sink node t through Ford-Fulkerson algorithm. **(10 Marks)**



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	3,4,5,6			10
Q2	3	3	3,4,5,6			10
Q3	3	3	3,4,5,6			10
Q4	4	4	3,4,5,6			10
Q5	4	4	3,4,5,6			10